

4(a)	any four from: I 'find optimum conditions for the enzyme mixture' A 'enzymes' for enzyme mixture 1 (find) suitable, raw material / crop wastes / substrate ; using Table 4.1 2 (find) crop waste that has high concentration of substrate for enzyme M ; ORA assume answer is about solution or mixture unless told about individual enzymes (M, N and O) 3 (find), optimum pH / optimum temperature ; A reference to, pH / temperature, for maximum activity I find optimum pH / temperature of each enzyme 4 (find) appropriate <u>concentrations</u> of the enzymes in the mixture ; I 'proportions' 5 <i>idea of</i> ratio of enzyme mixture to, raw material / substrate / crop waste ; 6 (find) the best pre-treatment of the, raw material / substrate / crop waste ; 7 (find) if any, inhibitors / cofactors, are <u>present in the solution</u> ; 8 (find) if any products are toxic ; 9 identify the sugars produced ; 10 (find) how long it takes to produce sugars / (find) rate of sugar production ; 11 <i>ref to</i> enzyme immobilisation ;	4
4(b)	any four from: 1 cellulose and, (named) protein / polypeptides, were present (in cell wall) ; e.g. expansins R if collagen 2 cellulose is broken down (fully) to β-glucose / β-glucose is the monomer of cellulose ; A 'made of' 3 protein / polypeptide / peptide, is broken down (fully) to amino acids ; A 'made of' 4 <i>idea that</i> partial hydrolysis has occurred to form, peptides / short chains of β-glucose ; 5 enzymes include cellulase and protease ; 6 breakage of glycosidic and peptide bonds ; A glucosidic 7 AVP ; e.g. beta 1-4 glycosidic linkage / exopeptidases producing amino acids / endopeptidases producing peptides	4